Societal Critical Coherence Index

A Toy Framework for Structural Stress and Constructive Transformation

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Abstract

We introduce a synthetic and ethically bounded toy framework for applying the Critical Coherence Index logic to social-system archetypes. The goal is not to rank real people, parties, countries, organisations, or communities, but to test whether structural diagnostics can separate raw activity, passive coherence, robustness, and constructive transformation.

The framework defines four primitive support variables: coherence H, balance B, raw social activity A_{raw}, and connectedness V. Raw activity is converted into controlled activity through an optimal-window response $S_{\text{eff}}(A)$, so that both stagnation and overdriven mobilisation are penalised. From these quantities we construct passive structural adherence $R_{\text{resp,soc}}$, activity-sensitive robustness $R_{\text{rob,soc}}$, active societal significance $R_{\text{sig,soc}}$, a societal structural-stress index CCI_{soc}, and a constructive-significance index ASI_{soc}.

A reproducible fixed-seed simulation evaluates six synthetic archetypes and a 5000-sample random ensemble. The highest constructive-significance score is obtained by the *creative democratic renewal* archetype ($\text{ASI}_{\text{soc}} \simeq 0.694$, $R_{\text{sig,soc}} \simeq 0.906$), while the highest structural stress is obtained by *polarized mobilisation* ($\text{CCI}_{\text{soc}} \simeq 1.018$). The result supports the distinction between low stress, high activity, and constructive transformation.

1 Purpose and Ethical Status

This paper is an experiment in structural analogy. It transfers the MAAT/CCI idea from physical and computational systems to a synthetic social-system toy model. It is deliberately not a social scoring system.

> *Social health is not the absence of conflict, but the capacity to transform conflict into coherent, balanced, and connected activity.*

The model must not be used to rank real people, parties, countries, organisations, populations, or communities. All examples are synthetic archetypes designed to test regime separation. The output should be interpreted as a conceptual diagnostic and not as a normative judgement.

2 From Physical CCI to Social-System Diagnostics

The original Critical Coherence Index was introduced as a structural diagnostic balancing activity and coherence in nonlinear systems. In the present paper we do not claim a literal physical derivation for social systems. Instead, we use the CCI logic as a controlled analogy for complex social states.

The central hypothesis is:

> *Activity is not automatically constructive, and stability is not automatically healthy. Constructive transformation requires controlled activity constrained by coherence, balance, and connectedness.*

3 Societal Support Sectors

The toy model uses four primitive support variables. The fifth quantity, robustness or respect, is not treated as an independent primitive sector. Instead, it is derived as an emergent closure from the primitive supports, consistent with the MAAT v1.2.1 convention.

Table 1: Primitive and derived social-system quantities in the toy model.

Symbol	Role	Social interpretation
H	primitive support	coherent orientation, shared sense-making
B	primitive support	balance between social tensions
A_{raw}	raw activity	mobilisation, reform pressure, civic activity
V	primitive support	trust, communication, connectedness
S_{eff}	derived support	controlled activity in an optimal window
$R_{\text{resp,soc}}$	derived closure	passive structural adherence
$R_{\text{rob,soc}}$	derived closure	activity-sensitive robustness
$R_{\text{sig,soc}}$	derived diagnostic	active constructive significance

4 Controlled Activity

Raw activity is not treated as intrinsically positive. Too little activity corresponds to stagnation, while too much activity may correspond to overdriven mobilisation, polarisation, or loss of integration. The toy model therefore defines a controlled-activity support:

$$S_{\text{eff}}(A_{\text{raw}}) = \exp\left[-\frac{1}{2}\left(\frac{A_{\text{raw}} - A_\star}{\sigma_A}\right)^2\right]. \tag{1}$$

The parameters used in the benchmark are

$$A_\star = 1.0, \qquad \sigma_A = 0.30. \tag{2}$$

Thus constructive activity is concentrated around an optimal transformation window rather than at maximal raw activity.

5 Societal Robustness and Active Significance

Following the emergent-closure logic, passive structural adherence is defined as

$$R_{\text{resp,soc}} = (HBV)^{1/3}. \tag{3}$$

The activity-sensitive robustness boundary is

$$R_{\text{rob,soc}} = \min\left[R_{\text{resp,soc}}, (HBS_{\text{eff}}V)^{1/4}\right]. \tag{4}$$

This says that a system cannot be more robust than its passive adherence, but its realised robustness is also limited by whether its activity is controlled.

Active societal significance is defined as

$$R_{\text{sig,soc}} = R_{\text{resp,soc}}^{1-\alpha} S_{\text{eff}}^{\alpha}, \qquad \alpha = 0.45. \tag{5}$$

Thus $R_{\text{sig,soc}}$ measures not whether a system merely satisfies its constraints, but whether its constraint-compatible structure becomes dynamically effective.

6 Societal CCI and Constructive-Significance Index

The societal CCI is used here as a structural-stress index:

$$\mathrm{CCI}_{\mathrm{soc}} = \frac{A_{\mathrm{raw}}}{H + B + V + R_{\mathrm{rob,soc}} + \varepsilon}. \tag{6}$$

Large values indicate that raw activity is high relative to coherence, balance, connectedness, and robustness.

We also define a constructive-significance index:

$$\mathrm{ASI}_{\mathrm{soc}} = \frac{R_{\mathrm{sig,soc}}}{1 + \mathrm{CCI}_{\mathrm{soc}}}. \tag{7}$$

This index is high only when active significance remains high while structural stress remains controlled.

7 Synthetic Experiment

The benchmark evaluates six synthetic archetypes:

- stagnant order,
- constructive reform,
- polarized mobilisation,
- fragmented activism,
- authoritarian stability,
- creative democratic renewal.

The support values are chosen phenomenologically to encode distinct toy regimes. They are not derived from empirical social data or from a microscopic social model.

The script also generates a fixed-seed random ensemble of 5000 synthetic states to visualise the phase space. The random ensemble is used only for qualitative regime structure, not for statistical inference.

8 Results

Table 2: Synthetic archetype results.

Archetype	S_{eff}	$R_{\mathrm{rob,soc}}$	$R_{\mathrm{sig,soc}}$	$\mathrm{CCI}_{\mathrm{soc}}$	$\mathrm{ASI}_{\mathrm{soc}}$
stagnant order	0.024	0.319	0.160	0.069	0.150
constructive reform	0.991	0.760	0.856	0.316	0.651
polarized mobilisation	0.375	0.345	0.358	1.018	0.178
fragmented activism	0.946	0.338	0.537	0.787	0.300
authoritarian stability	0.040	0.228	0.144	0.155	0.125
creative democratic renewal	0.998	0.836	0.906	0.305	0.694

The highest $\mathrm{ASI}_{\mathrm{soc}}$ and $R_{\mathrm{sig,soc}}$ are obtained by the creative democratic renewal archetype. The highest $\mathrm{CCI}_{\mathrm{soc}}$ is obtained by polarized mobilisation. This separates constructive activity from raw stress.

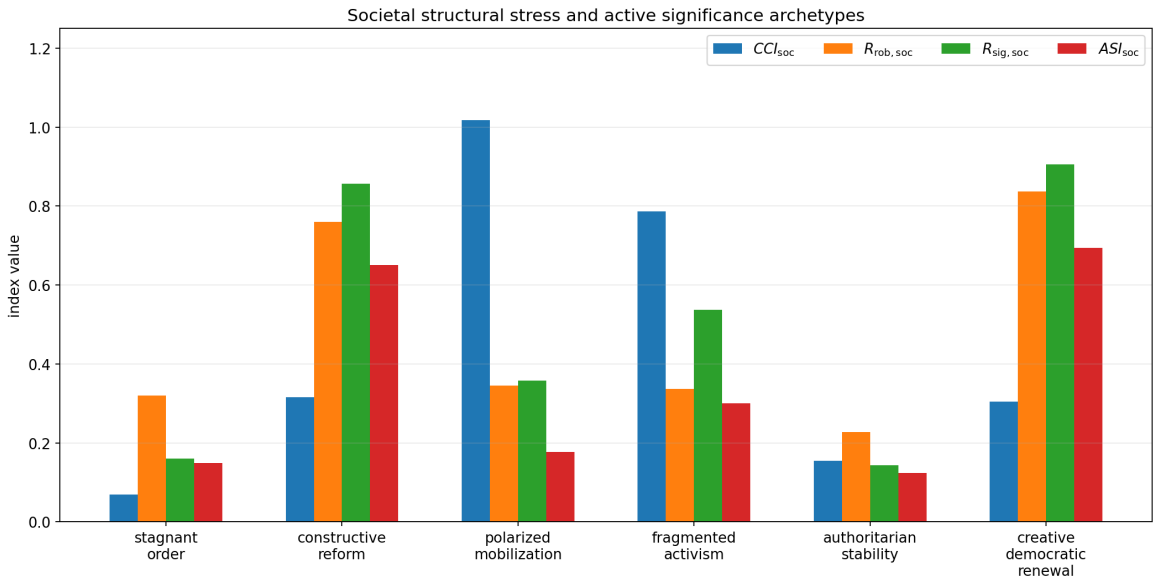

Figure 1: Synthetic archetype comparison. Constructive reform and creative democratic renewal reach high active significance without maximal structural stress.

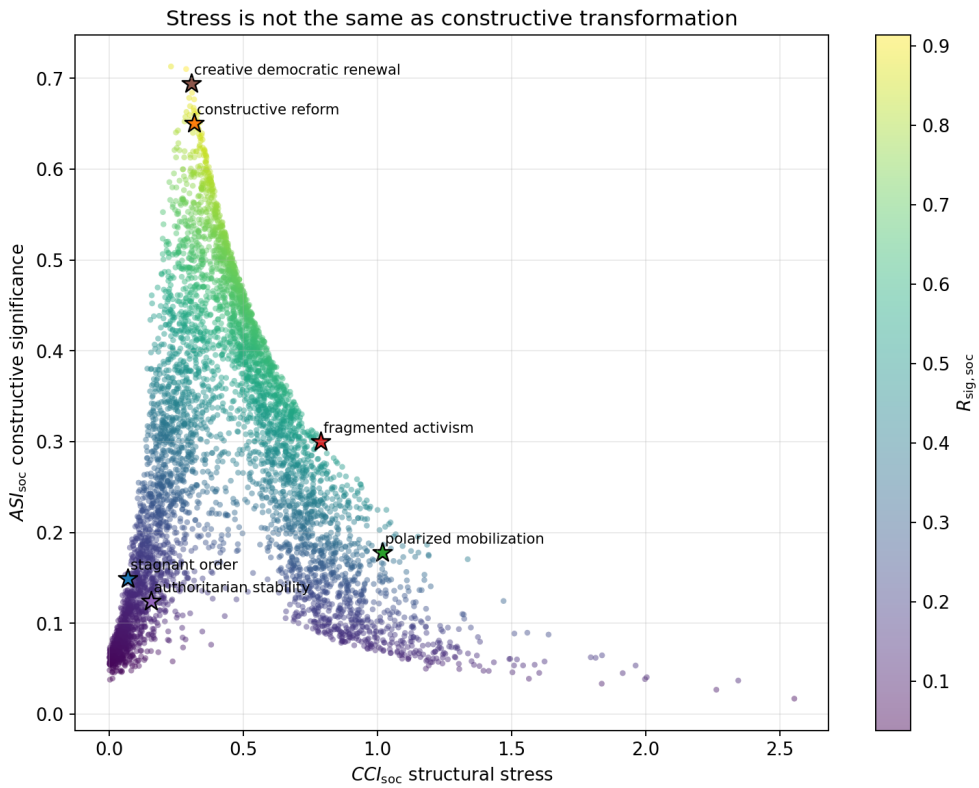

Figure 2: Synthetic phase-space view. Structural stress and constructive significance are not equivalent. High activity can be polarising, while low stress can be stagnant.

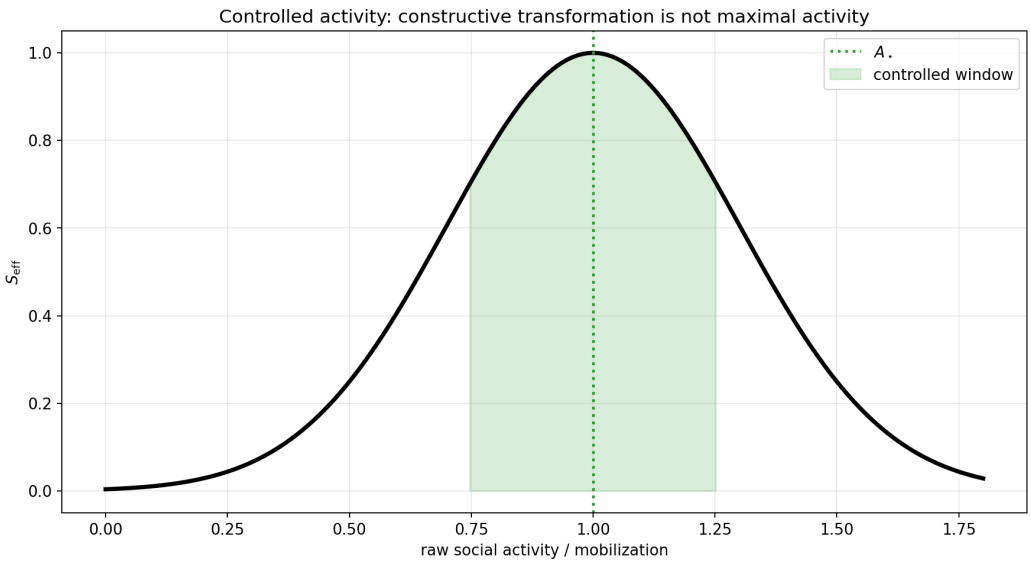

Figure 3: Controlled activity window. Constructive transformation is modelled as activity near an optimal window, not as maximal raw activity.

In the 5000-sample synthetic ensemble, the largest regime is mixed, followed by constrained stability and polarization-risk states:

Table 3: Synthetic ensemble regime counts, fixed seed 44.

Regime	**Count**
mixed regime	2876
constrained stability	784
polarization risk	672
fragmentation risk	379
constructive transformation	221
stable but stagnant	68

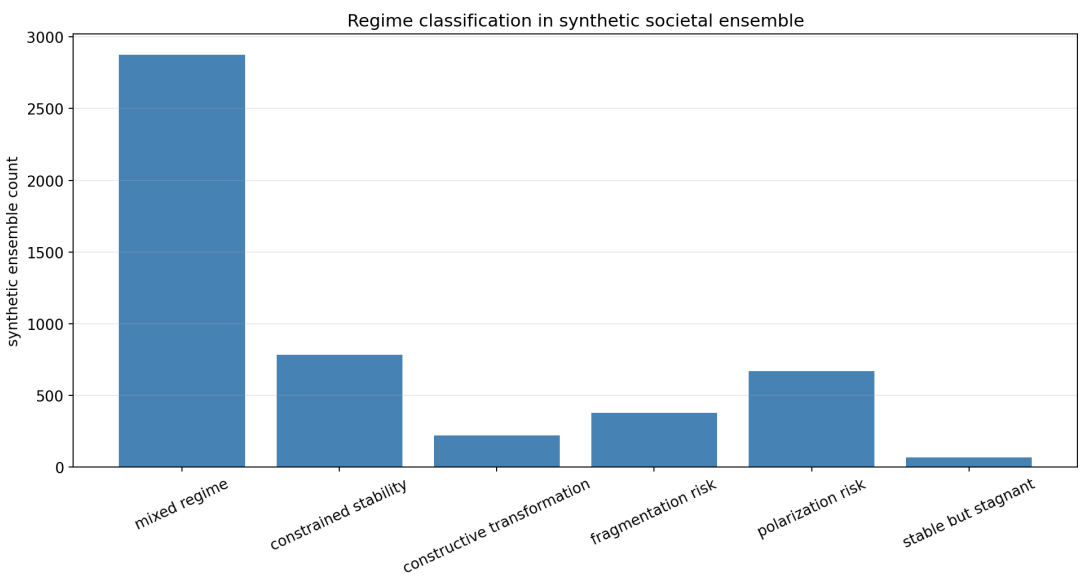

Figure 4: Regime counts in the synthetic random ensemble.

9 Interpretation

The toy model demonstrates three separations:

- Low stress is not sufficient for constructive transformation.
- High activity is not sufficient for social significance.
- Active significance requires controlled activity supported by coherence, balance, and connectedness.

The stagnant-order archetype has low structural stress ($\mathrm{CCI}_{\mathrm{soc}} \simeq 0.069$) but also low constructive significance ($\mathrm{ASI}_{\mathrm{soc}} \simeq 0.150$). The polarized-mobilisation archetype has the highest stress ($\mathrm{CCI}_{\mathrm{soc}} \simeq 1.018$) but low constructive significance ($\mathrm{ASI}_{\mathrm{soc}} \simeq 0.178$). The creative-democratic-renewal archetype has high controlled activity, high robustness, and the highest constructive significance ($\mathrm{ASI}_{\mathrm{soc}} \simeq 0.694$).

10 Limitations

This framework is intentionally limited:

- The archetypes are synthetic and do not represent real societies.
- The support values are phenomenological.
- The regime classifier is heuristic.
- No causal claims are made.
- No empirical political, cultural, demographic, or institutional data are used.
- The model should not be used for governance, surveillance, persuasion, or political scoring.

11 Reproducibility

The experiment is stored in the repository under:

`experiments/societal_cci/`

Run:

```
python3 societal_cci_toy.py
```

The script writes:

- `outputs/societal_cci_results.json`,
- `outputs/societal_cci_archetypes.csv`,
- `outputs/societal_cci_ensemble.csv`,
- four PNG figures used in this paper.

The public research repository is:

https://github.com/Chris4081/structural-selection-principle

12 Conclusion

We introduced a synthetic Societal Critical Coherence Index as an ethical toy extension of the MAAT/CCI framework. The model distinguishes raw activity from controlled activity, passive adherence from active significance, and low stress from constructive transformation.

The main conceptual result is that social structure cannot be evaluated by stability or activity alone. In this toy framework, constructive transformation emerges only when activity is held inside a coherent, balanced, and connected structural envelope.

References

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